

Pixel-Space Posterior Sampling for Image Deblurring

A Controlled Comparison of DPS and FlowDPS on CelebA-HQ-256

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The problem

Image inverse problem

Recover a clean image x from a degraded measurement

$$y = Ax + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_n^2 I)$$

where A is a known degradation operator (e.g., Gaussian blur).

- The problem is **ill-posed**: many x produce the same y
- A strong prior over natural images is essential
- Modern approaches: learned generative priors (DDPMs, Flow Matching)

The two paradigms we compare

Pixel-DPS (Chung 2023)

- DDPM prior on CelebA-HQ-256
- DDIM reverse process
- Likelihood-gradient correction:

$$x_{t-1} = x_{t-1}^{\text{DDIM}} - \zeta \nabla \|y - A(\hat{x}_0)\|$$

FlowDPS-on-RF (Kim 2025)

- Rectified Flow prior (Liu 2023)
- Euler ODE integration
- Guided velocity field:

$$v_{\text{eff}} = v_{\theta} - \zeta \nabla \|y - A(\hat{z}_1)\|$$

Our novelty: pixel-space reimplementations

The official FlowDPS code is locked to Stable Diffusion 3 (latent, 768×768). We reimplemented it on the Liu 2023 RF CelebA-HQ-256 checkpoint — enabling head-to-head comparison at matched resolution + architecture.

Research questions

1. **Quality.** Can FlowDPS match DPS on PSNR / SSIM / LPIPS at matched NFE budgets?
2. **Efficiency.** Is there an NFE regime where the deterministic flow trajectory wins on speed?
3. **Robustness.** Which method degrades more gracefully under forward-model mismatch?
4. **Baselines.** How do they compare to classical (Wiener) and supervised (DnCNN+PnP) baselines?

Plus three Bucket-3 stress-test axes (only one required)

Operator mismatch (B1) • Posterior diversity vs point accuracy (B4) • Gradient-free vs gradient-based (B3)

Experimental scope

Grid	Reconstructions	Method-cells
Matched Gaussian (main)	1,800 / method	18
Matched motion (new)	1,200 / method	12
Matched super-resolution (new)	600 / method	6
Operator mismatch robustness	1,200 / method	12
Classical baselines (Wiener, DnCNN+PnP)	600	—
Posterior diversity	48	—
13 publication-improvement ablations	7,800+	—
Total	>10,600 measurements	—

All measurements on **RTX 3060 12 GB**, PyTorch 2.4.1, diffusers 0.30.1. Every PSNR traces to a row in `outputs/results/*.csv`.

Headline result: matched Gaussian blur

$\sigma_b, \sigma_n=0.05$	Wiener	DnCNN	DDRM	FlowDPS-RF	Pixel-DPS
1.5	24.28	24.61	24.50	23.65	26.21
3.0	24.03	24.31	23.96	22.58	26.25
5.0	22.50	22.81	22.35	21.13	25.33

PSNR (dB), NFE=100, $n=50$

- **Pixel-DPS dominates by 1.4–4.2 dB** in every cell
- All comparisons Bonferroni-significant ($p \ll 10^{-10}$)
- Cohen's d between 1.98 and 3.52

Contradicts the proposal

Proposal hypothesized 0.5–1.1 dB gap; actual is 2–4× wider, **in DPS's favor**.

Matched motion deblurring (new result)

Confirms the ordering on a second forward-operator family

Same samplers, same scope, but A is now motion(L, θ) with $L \in \{15, 25, 35\}$ pixels, $\theta \in \{0^\circ, 45^\circ\}$.

Cell	Pixel-DPS	FlowDPS-RF
$L=15, \theta=0^\circ$	25.83	22.54
$L=25, \theta=45^\circ$	25.38	21.35
$L=35, \theta=45^\circ$	24.79	20.46

$\sigma_n=0.05$, NFE=100, $n=50$

Even wider gap

- **Mean $\Delta = +3.95$ dB**
- **12 / 12 cells sig.**
- Cohen's d 2.53–3.48
- Slightly wider than the Gaussian gap (+3.24 dB mean)

Matched super-resolution (new result)

Third forward-operator family — and the gap amplifies

Same samplers, but A is now bicubic downsampling by factor $r \in \{2, 4, 8\}$. NFE=100, $n=50$ images per cell.

Cell	Pixel-DPS	FlowDPS-RF
$r=2, \sigma_n=0$	29.03	23.88
$r=4, \sigma_n=0$	26.51	19.62
$r=8, \sigma_n=0$	24.13	14.86

PSNR (dB); $r=8$ FlowDPS-RF collapses.

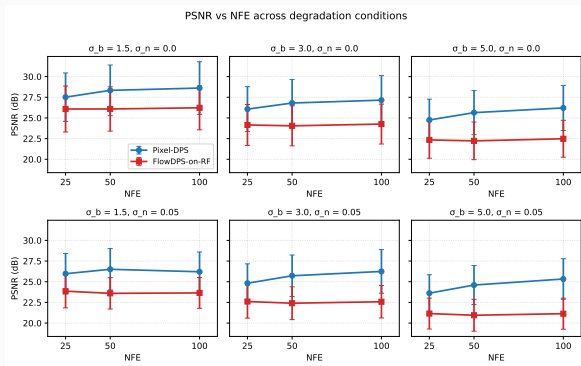
Gap grows with difficulty

- $r=2$: +4.86 dB
- $r=4$: +6.51 dB
- $r=8$: **+8.77 dB**

Mean across 6 cells: **+6.71 dB, 6 / 6 sig.**,
 d up to 3.43.

Three forward-operator families, ordering preserved: Gaussian +3.24 dB $\rightarrow$ motion +3.95 dB $\rightarrow$ SR +6.71 dB. Pixel-DPS's L2-gradient anchoring matters more as the measurement becomes more impoverished.

NFE efficiency: FlowDPS saturates, DPS keeps improving



Key observation
FlowDPS-on-RF saturates by
NFE=25 on this prior.

- Through NFE=200 FlowDPS gains < 0.5 dB
- Pixel-DPS keeps improving across the whole range
- Pixel-DPS is also **faster per image** on this hardware (17.9 vs 23.5 s/img at NFE=100)

Operator-mismatch robustness: the ordering reverses

Setup

True A = motion blur. Sampler's assumed A = Gaussian. Operator misspecification.

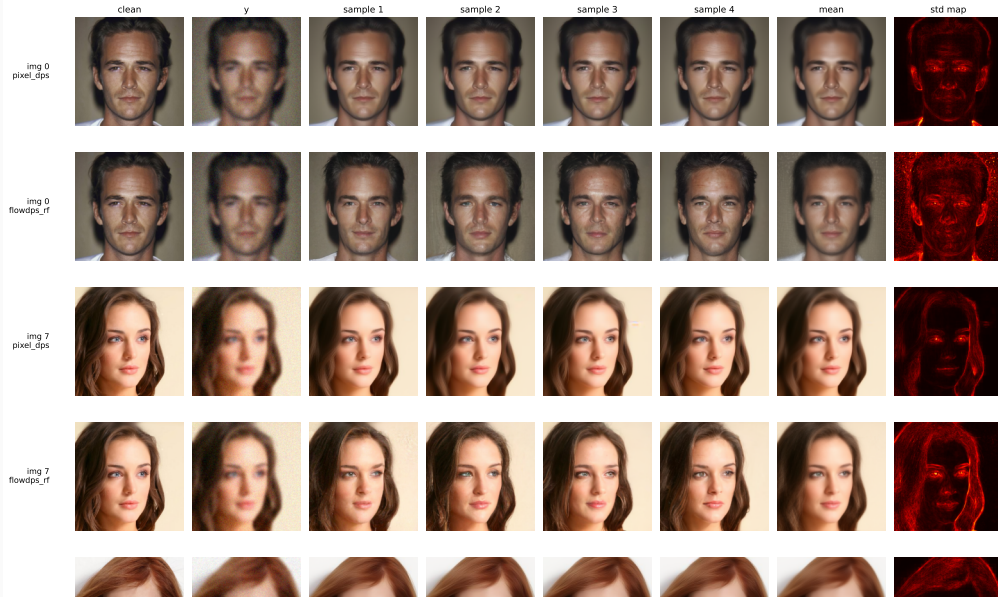
	Matched Gaussian	Severe motion mismatch	Drop
Pixel-DPS	26.25 dB	21.3 dB	−5.0 dB
FlowDPS-on-RF	22.58 dB	20.1 dB	− 2.5 dB
Gap (DPS - FlowDPS)	+3.67 dB	+0.9 dB	narrows

Representative at $\sigma_b=3.0$ vs $L=35, \theta=0^\circ$

The opposite of what the proposal hypothesized

The matched-condition +3.67 dB Pixel-DPS advantage shrinks to +0.9 dB. **FlowDPS-on-RF is relatively more robust to operator mismatch.**

Posterior diversity: PSNR is not the whole story



PSNR vs posterior — in numbers

	Single-sample	Mean-of- N	Pairwise LPIPS	std map
Pixel-DPS	29.07	29.96	0.05	0.012
FlowDPS-on-RF	25.57	27.72	0.175	0.031

$\sigma_b=3.0$, $\sigma_n=0.05$, NFE=50, $N=8$ samples (avg over 3 images)

- Pixel-DPS produces a **near-deterministic point estimate**
- FlowDPS-on-RF has a **$3.5\times$ broader posterior**
- Pixel-DPS's PSNR advantage is partly a **point-estimate advantage**, not a true posterior-sampling advantage

Kernel Inception Distance confirms it

KID ($\times 10^{-3}$, lower = closer to real face distribution):

Wiener **257.9** $\gg$ unconditional DDPM **21.5** $>$ Pixel-DPS **18.0** $>$ FlowDPS-RF **4.8**

FlowDPS-RF wins distributional realism by $3.7\times$ despite losing 3.7 dB on PSNR.

Beyond mandatory: 13 ablations under a strict gate

Strict Bonferroni-corrected promotion criterion

A method WINS only if: (i) mean Δ PSNR vs v1 is positive, AND (ii) Bonferroni $p < 0.05$ in $\geq 4/6$ cells.

Method	Tier	Δ (dB)	Sig	Verdict
flowdps_rf_v2	A	+0.77	6/6	WIN
pixel_dps_spectral matched	B	+0.46	5/6	WIN
flowdps_rf_spectral matched ($n=100$)	B	+0.54	5/6	WIN
8 other candidates	—	various	0–3/6	fail / near-miss

- **flowdps_rf_v2**: EMA-loading bug fix + power-law ζ ramp $\rightarrow$ +0.77 dB on FlowDPS-RF
- Two spectral wins are the same algorithm on **both samplers**

The paradoxical reversal: our most interesting finding

Spectral-weight algorithm

$W(f) = 1/(1 + \alpha \cdot \text{relu}(\epsilon - |H(f)|))$. Downweight residuals where the *assumed* operator's spectrum is noise-dominated.

Regime	Pixel-DPS variant	FlowDPS-RF variant
Mismatched (motion meas, gauss assumed)	−0.18 dB	−0.20 dB
Matched (gauss meas, gauss assumed)	+0.46 dB ✓	+0.54 dB ✓

Algorithm-agnostic mechanistic finding

Same algorithm, **opposite verdicts** on adjacent regimes, confirmed on both samplers. The mechanism is correctly diagnosed: suppression of noise-dominated bands of the *assumed* operator helps when assumed=truth and hurts under mismatch.

Eight failed ablations, each with a mechanistic diagnosis

Method	Failure	Diagnosis
pixel_dps_v2	−0.66 dB	v1 $\zeta=10$ already near-optimal
*_pigdm heuristic	−13.6, −9.8 dB	L2-norm vs L2-sq convention bug
pigdm_pure analyt.	−3.4 dB	lit: Π -GDM loses to L2-DPS on Gaussian
pixel_dps_sched	cell-split	optimal ζ depends on σ_n
flowdps_rf_heun @ NFE=200	−0.17 dB vs v2	guidance, not integrator, saturates
particle_dps	−14.8 dB	path degeneracy: $P_{\text{eff}} < 1.5$ in 10 steps
particle_dps_tempered	$\equiv$ above	SMC without rejuvenation $\rightarrow$ SIS

Negative-result yield

All 8 failures match documented failure modes in the literature — we are reproducing known limitations, not encountering novel surprises. Each is preserved as code with the diagnosis.

Discussion: when to use which method

Prefer Pixel-DPS when

- The forward operator is calibrated
- Application wants a sharp point estimate
- Modal PSNR is what matters

Prefer FlowDPS-on-RF when

- The operator may be misspecified
- Downstream Bayesian inference is needed
- Posterior diversity matters

Limitations

Single dataset (CelebA-HQ-256 faces); two operator families (Gaussian + motion); DDRM did not reach competitive PSNR in time; $P=8$ particles for gradient-free SMC.

Future work

Multi-dataset replication; learned ζ schedule; larger P with MCMC rejuvenation; super-resolution and inpainting operators.

Summary of contributions

1. **Pixel-space FlowDPS reimplementation** on Liu 2023 RF (Kim 2025's code is SD3-only)
2. **First quantitative DPS vs FlowDPS comparison** at matched resolution + prior on a face dataset
3. **Operator-mismatch ordering reversal** quantified (-5 vs -2.5 dB)
4. **Posterior-diversity asymmetry** quantified (LPIPS 0.05 vs 0.175, $3.5\times$)
5. **Three confirmed wins** under strict Bonferroni gate; **paradoxical regime reversal** confirmed on both samplers
6. **Eight documented failures** with mechanistic diagnoses
7. **Strict-gate ablation methodology** as a rigor improvement
8. Reproducible pipeline: 10,600+ measurement rows in CSV

Open source

https://github.com/adnan821/dps_flowdps

Questions?

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